EEE4114F Class Test 1

March 2025

Duration: 1 hour

Total: 20 marks

Empl ID: ______________________________

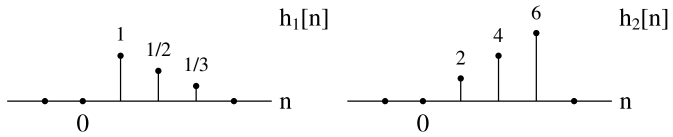
Instructions:

- This is a closed-book test.
- There are *four* questions totalling 20 marks. You must answer all of them.
- Questions should be answered in the space provided on the test script. If necessary, you may also write on the backs of the printed pages, but please indicate where you have done so.
- You must use a pen to complete the exam – pencil will only be marked for drawings. Values indicated on drawings must be written in pen.
- Please show all of your working clearly – your method and approach matter.
- Do not write your student number or name anywhere on this script; instead, provide only the 7-digit EMPLID found on your student card.
- A formula sheet is provided at the end of the paper; you may detach this, but please do so carefully so as not to damage the rest of the script.
- You have 1 hour.

Question	Marks	Grade
1	5	
2	5	
3	5	
4	5	
Total:	20	

Question 1 [5 marks]

You are given a composite system that is formed by cascading two LTI processors with the following impulse responses:



Sketch the overall impulse response $h[n]$ for the composite system, and provide an analytical expression for it.

Question 2 [5 marks]

Consider an LTI system represented by the difference equation

$$y[n] = x[n+1] - x[n]$$

where $x[n]$ and $y[n]$ are the input and output signals respectively.

1. Determine the impulse response $h[n]$ of the system, and thus determine whether the system is causal.

2. Find the step response of the system.

3. Find the frequency response $H(e^{j\omega})$ and plot its magnitude over the range $-2\pi \leq \omega \leq 2\pi$.

Question 3 [5 marks]

An LTI system function is provided below.

$$H(z) = \frac{8z-4}{8z-2}, \qquad |z| > \frac{1}{4}.$$

Find the impulse response of an appropriate *causal* inverse system, and then comment on the stability of this inverse system.

Question 4 [5 marks]

You are given the system equation below with ROC $|z| > 0.5$:

$$H(z) = \frac{z^2}{1 + 0.25z^2}$$

NOTE: This question was not marked due to an error in the printed copies; denominator should have been ($z^2 + 0.25$) but was given to students as ($1 + 0.25z^2$) instead. Test marked out of 15.

1. Find the system's impulse response.

2. Sketch the system's pole-zero plot in the z-plane, and indicate the ROC.

EEE4114F Information Sheet

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}dt \quad \Longleftrightarrow \quad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t}d\Omega \qquad \text{(CTFT)}$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega \qquad \text{(DTFT)}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi j}\oint_C X(z)z^{n-1}dz \qquad \text{(ZT)}$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \Longleftrightarrow \quad x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with} \quad W_N = e^{-j\frac{2\pi}{N}} \qquad \text{(DFT)}$$

Discrete-time Fourier transform properties

Sequences $x[n]$, $y[n]$	Transforms $X(e^{j\omega})$, $Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d}X(e^{j\omega})$	Time shift
$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega-\omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
$1 \quad (-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (\lvert a\rvert < 1)$	$\frac{1}{1-ae^{-j\omega}}$
$u[n]$	$\frac{1}{1-e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n+1)a^n u[n] \quad (\lvert a\rvert < 1)$	$\frac{1}{(1-ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \lvert\omega\rvert < \omega_c \\ 0 & \omega_c < \lvert\omega\rvert \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)}e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n]$, $y[n]$	Transforms $X(z)$, $Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d}X(z)$	ROC $= R_x$	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC $= \lvert z_0\rvert R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC $= \frac{1}{R_x}$	Time reversal
$nx[n]$	$-z\frac{dX(z)}{dz}$	ROC $= R_x$	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC $= R_x$	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert < \lvert a\rvert$
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$\lvert z\rvert > 0$
$\cos(\omega_0 n)u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
$r^n\cos(\omega_0 n)u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > \lvert r\rvert$

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for} \quad a \neq 1.$$